

Address Spaces

1 Core Idea

Definition

An **address space** is the set of memory addresses that a process is allowed to use.

The address space abstraction gives each process the illusion of private memory.

Big Idea

process address $\neq$ physical address

Why It Matters

Address spaces solve two problems that direct physical addressing cannot handle well: **protection** and **relocation**.

2 Problem With Physical Memory

If programs can use physical memory directly, several things can go wrong.

Problem	Result
User program can address all memory	It may overwrite the operating system.
Two programs use the same address	They may overwrite each other's code or data.
Program loaded in a different place	Absolute jumps such as <code>JMP 28</code> may go to the wrong physical address.

Main Issue

Physical memory exposes real machine addresses. Address spaces hide those addresses behind a safer process-level view.

3 Protection and Relocation

Need	Question answered
Protection	Is this process allowed to access this address?
Relocation	Which physical address should this process address map to?

Protection alone is not enough. IBM 360-style protection keys could stop illegal access, but they did not automatically turn Program *B*'s address 28 into the correct physical address 16,412.

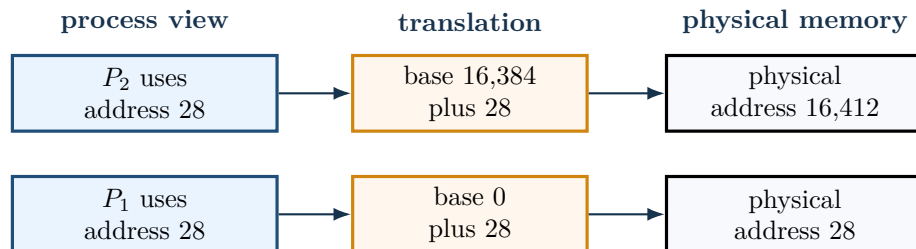
4 Address Space Examples

An address space is any set of valid addresses or names.

System	Address space
Process memory	Addresses a process may use.
Telephone numbers	All valid phone numbers in a numbering system.
x86 I/O ports	0 to 16,383.
IPv4	0 to $2^{32} - 1$, with reserved ranges.
.com domains	Valid domain-name strings ending in .com.

5 Private Address Spaces

Each process can use the same address values, but those values can map to different physical memory locations.



Interpretation

The address value 28 is not globally fixed. Its physical meaning depends on which process is running.

6 Base and Limit Registers

Mechanism

A simple way to implement address spaces is to use two CPU registers: **base** and **limit**.

Register	Meaning
Base	Starting physical address of the process.
Limit	Size of the process's legal address range.

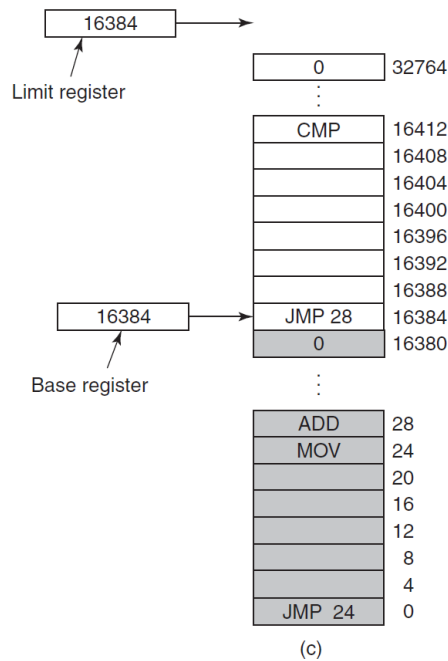


Figure 3-3. Base and limit registers can be used to give each process a separate address space.

Figure Interpretation

For the second program, the base register contains 16,384. A local address such as 28 is translated to physical address $16,384 + 28 = 16,412$. The limit register checks that the local address is inside the process's allowed range.

When the operating system runs a process, it loads that process's base and limit values into the CPU. User programs normally cannot change these registers.

7 Translation Rule

Let:

- A = address generated by the process;
- B = base register value;
- L = limit register value.

Base-Limit Translation

$$\begin{aligned} A < L &\Rightarrow \text{physical address} = B + A \\ A \geq L &\Rightarrow \text{fault} \end{aligned}$$

Two Jobs at Once

The limit check gives protection. The base addition gives relocation.

8 Translation Examples

Setup

Program B starts at physical address 16,384 and has length 16 KB.

$$B = 16,384, \quad L = 16,384$$

Legal address: JMP 28

Program B executes:

JMP 28

Step	Result
Generated address	$A = 28$
Limit check	$28 < 16,384$, so the access is legal.
Translation	$16,384 + 28 = 16,412$

Result

JMP 28 lands at physical address 16,412, which is Program B 's own instruction.

Illegal address: $A = 20,000$

Step	Result
Generated address	$A = 20,000$
Limit check	$20,000 \geq 16,384$, so the address is outside the process.
Translation	No physical address is produced. The CPU raises a fault.

Result

The access is aborted before memory is touched.

9 Dynamic Relocation

Definition

Dynamic relocation means addresses are translated while the program runs.

This avoids changing the program file while loading. The hardware adds the base register on every instruction fetch and data access.

Method	Difference
Static relocation	Loader modifies address constants before execution.
Dynamic relocation	CPU translates addresses during execution.

10 Strengths and Weaknesses

Point	Explanation
Strength	Simple private address spaces.
Strength	Protects against out-of-range memory access.
Strength	Program can be loaded wherever there is room.
Weakness	Every memory reference needs one comparison and one addition.
Weakness	Each process usually needs one contiguous block of physical memory.

Hardware Cost

Base-limit relocation is simple, but the CPU must check and translate every memory reference.

11 Quick Comparison

Concept	Meaning
Physical memory only	Program addresses are real physical addresses.
Address space	Process has its own usable address range.
Base register	Where the process starts in physical memory.
Limit register	How far the process may address.
Protection	Prevent access outside the process.
Relocation	Convert process address to physical address.